

Internship Project Plan

After reviewing the project reports of the previous interns and exploring different implementation possibilities, I would like to work on the following projects during my internship.

Project 1: Thermal Power Plant Monitoring & Fault Detection Platform

Project Description

This project aims to develop a web-based platform for monitoring thermal power plants and detecting operational faults in generation units using Artificial Intelligence.

The platform will allow administrators to manage thermal power plants, generation units, and engineering personnel through a centralized dashboard. The dashboard will display the total number of plants, total units, running units, units under maintenance, faulty units, and plant-wise operational statistics.

Each plant will have a dedicated monitoring page showing localized metrics such as unit health, power generation, fault counts, maintenance requests, and performance trends. Engineers will be able to access a dedicated Fault Detection module where they can input operational telemetry values such as Turbine RPM, Steam Temperature, and Pressure.

A trained machine learning model will analyze the operational parameters and classify the unit condition as either Normal or a specific fault type such as Blade Deformation, Turbine Inefficiency, Pressure Anomaly, or Overheating. The system will display the predicted fault class along with a confidence score.

If a fault is detected with a confidence score above a predefined threshold, the system will automatically update the faulty unit count for both the plant and the global dashboard. Engineers will be able to raise maintenance requests with different priority levels, which can later be managed by administrators.

The platform will also include an AI-powered Plant Assistant that allows users to interact with plant data using natural language. Users can ask questions regarding thermal plant machines, their use, plant performance, fault trends, maintenance history, unit health, and operational statistics. The assistant will provide insights from dashboard data, and help users quickly retrieve information without manually navigating through multiple dashboards.

Technologies

Frontend

- React.js
- Vite
- CSS
- Recharts

Backend

- Python
- Flask / FastAPI

Database

- MySQL
- SQLAlchemy ORM

Authentication & Security

- JWT Authentication

Machine Learning

- Scikit-Learn
- Random Forest / XGBoost

AI Integration

- Gemini/OpenAI API

Key Features

- Thermal power plant management
 - Generation unit management
 - Engineer management
 - Role-based authentication and authorization
 - Plant-wise monitoring dashboards
 - Real-time KPI tracking
 - AI-powered fault detection
 - Fault history management
 - Maintenance request generation
 - Request management hub
 - AI-powered Plant Assistant
 - Plant and unit analytics dashboard
 - Power generation tracking
 - Unit performance monitoring
 - Admin asset management tools
 - Responsive industrial-themed UI
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Development Timeline (6 Weeks)

Week 1

- Got introduced to managers, mentors, and team members.
- Understood the company's business operations and organizational structure.
- Learned about the Thermal Power Business Unit and department responsibilities.
- Reviewed project reports and work completed by previous interns.
- Analyzed different project ideas and evaluated their feasibility.
- Studied existing thermal plant monitoring systems and industrial dashboards.
- Researched technologies and tools required for implementation.
- Finalized project scope, architecture, and development roadmap.
- Prepared project description document and implementation plan.
- Created a week-wise internship timeline.

Week 2

- Design the database schema and overall system architecture for the Thermal Power Plant Monitoring & Fault Detection Platform.
 - Design UI wireframes and dashboard layouts.
 - Begin frontend development using React.js.
 - Implement global dashboard and plant dashboard interfaces.
 - Set up backend project structure and database connectivity.
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Week 3

- Develop backend APIs for plant, unit, and engineer management.
 - Implement JWT-based authentication and authorization.
 - Integrate frontend and backend services.
 - Implement plant-wise and unit-wise monitoring modules.
 - Develop dashboard modules for displaying operational statistics and KPIs.
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Week 4

- download and clean the kaggle dataset for machine learning.
 - Train and integrate the fault detection model.
 - Store prediction history and fault records in the database.
 - Develop fault analytics dashboards and visualizations.
 - Implement automated fault status updates.
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Week 5

- Implement maintenance request generation workflow.
- Add request sorting and filtering functionality.
- Implement admin controls for plant, unit, and engineer management.
- Complete analytics dashboards and reporting modules.

- Perform integration testing and bug fixing.
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Week 6

- Refine UI/UX and optimize application performance.
 - Conduct end-to-end testing and validation.
 - Integrate LLM API and implement the AI Plant Assistant.
 - Enable natural language querying of plant analytics and operational data.
 - Deploy the ThermalWatch platform for demonstration.
 - Prepare project documentation and testing reports.
 - Demonstrate the completed project to mentor.
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NOTE

Along with the project mentioned in the timeline above, I will also be working on the Carbon Footprint Calculator & Sustainability Assistant in parallel. If I am able to complete it before or by the end of my internship, I will present it as an additional project.

Additionally, in the unlikely event that I am unable to complete the Thermal Plant Monitoring & Fault Detection project within the internship timeline, I will present the Carbon Footprint Calculator & Sustainability Assistant as my primary project deliverable.

Project 2 : Carbon Footprint Calculator & Sustainability Assistant

Project Description

This project aims to build a web platform that helps users calculate, track, and reduce their carbon footprint.

The website will provide educational content explaining carbon footprint, its environmental impact, and methods to reduce emissions. Users will be able to calculate emissions from household electricity usage, personal vehicles, flights, and public transportation through dedicated calculators.

An AI-powered chatbot will assist users by answering sustainability-related questions, explaining carbon offsetting concepts, and helping users estimate their emissions conversationally.

The system will also generate personalized recommendations and suggest alternative choices that can help reduce carbon emissions.

Technologies • React.js • MySQL • Gemini / OpenAI API • JWT Authentication	Key Features • Carbon footprint calculators • Sustainability score generation • AI-powered chatbot • Personalized recommendations
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